PES UNIVERSITY
100 feet Ring Road, BSK 3rd Stage
Bengaluru 560085

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UE23CS342BA1
SUPPLY CHAIN MANAGEMENT (SCM)
For Engineers

PROJECT REPORT
on
<Problem No. PS1>

<- Order & Inventory Visibility>

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Submitted by: Team Code#: 234-634-663

SRN 1 PES1UG23CS234	Name 1 Harsh Raj	6<D >
SRN 2 PES1UG23CS634	Name 2 Tanisha Mangal	6<K>

SRN 3 PES1UG23CS663	Name 3 V. Pooja Reddy	6<K>

_Class of _Prof. Raghu B. A._

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Introduction

In today's fast-paced business environment, efficient supply chain management plays a critical role in ensuring timely delivery of products and maintaining optimal inventory levels. Organizations must continuously monitor orders, track shipments, and manage stock availability to meet customer demands and avoid operational inefficiencies.

However, lack of visibility across different stages of the supply chain often leads to issues such as delayed deliveries, stock shortages, and poor demand planning. To overcome these challenges, businesses rely on data-driven solutions that provide real-time insights and predictive capabilities.

This project aims to develop an integrated system for Order and Inventory Visibility, supported by an interactive dashboard and basic machine learning models. The system enables monitoring of order status, analysis of inventory levels, and identification of shipment delays. Additionally, it incorporates demand forecasting techniques to predict future order trends.

By combining data visualization with predictive analytics, the project demonstrates how modern tools can enhance decision-making, improve operational efficiency, and support better planning in supply chain management.

Problem No. & Description

Problem No.: 1

Description:

Supply chain operations involve multiple interconnected processes, including order processing, inventory management, and shipment tracking. In many cases, these processes are handled independently, resulting in limited visibility and delayed decision-making.

The primary problem addressed in this project is the **lack of a unified system** to monitor and analyze key supply chain components. Without proper insights:

- Businesses may face unexpected delays in deliveries
- Inventory levels may become unbalanced (overstock or stockouts)
- Demand patterns may not be properly understood

To address these challenges, this project proposes a solution that:

- Tracks orders based on their status
- Monitors inventory availability across products
- Identifies delays in shipment
- Uses machine learning to forecast future demand

The goal is to create a system that provides **clear visibility, actionable insights, and predictive capabilities**, enabling better management of supply chain operations.

Scope

The scope of this project is to design and implement a data-driven system that enhances visibility across key components of supply chain operations. The system focuses on integrating order data, inventory information, and shipment details into a unified dashboard for analysis and decision-making.

The project covers the following areas:

- Monitoring order status across different stages such as Delivered, Delayed, Packed, and In Transit
- Tracking inventory levels for multiple products across warehouses
- Identifying shipment delays and analyzing delivery performance
- Visualizing data through an interactive dashboard using Power BI
- Applying machine learning techniques to forecast future demand and estimate delays

The dataset used in this project is a simulated dataset that represents real-world supply chain scenarios. The solution is designed for demonstration and analytical purposes and can be extended to real-time systems in practical applications.

Objectives

The primary objectives of this project are:

- To develop an interactive dashboard for real-time monitoring of supply chain operations
- To provide clear visibility into order status, inventory levels, and shipment delays
- To identify inefficiencies such as delayed deliveries and low stock conditions
- To enable data-driven decision-making through visual insights and key performance indicators
- To implement machine learning models for predicting future order demand and shipment delays
- To demonstrate how predictive analytics can improve planning and operational efficiency

Solution with Key Features

The proposed solution is an integrated system that combines data visualization and machine learning to improve visibility and efficiency in supply chain operations. The system is implemented using Power BI for dashboard development and Python-based machine learning models for predictive analysis.

The solution is designed to provide real-time insights into orders, inventory, and shipment performance, enabling supply chain managers to monitor operations effectively and make informed decisions.

Key Features

- **Interactive Dashboard**
  A user-friendly Power BI dashboard that visually represents supply chain data through charts, graphs, and KPI indicators.
- **Order Tracking System**
- Tracks orders based on their current status such as Delivered, Delayed, Packed, Shipped, and In Transit, helping identify bottlenecks in the order lifecycle.
- **Inventory Monitoring**
  Displays stock levels for different products, enabling identification of low-stock and overstock situations.
- **Shipment Delay Analysis**
  Highlights delays in deliveries by calculating delay days for each order, helping improve logistics performance.
- **Warehouse-Based Filtering**
  Allows users to filter data based on warehouse location (e.g., Bangalore, Chennai), enabling localized analysis.
- **Trend Analysis**
  Visualizes order trends over time, helping in understanding demand patterns.
- **Machine Learning Integration**
  Incorporates predictive models to:
- Forecast future order demand

- Predict shipment delays
- This integrated approach ensures both descriptive (what is happening) and predictive (what will happen) insights.

KPIs Tracked

The system tracks several Key Performance Indicators (KPIs) to evaluate the performance of supply chain operations:

- **Total Orders**
  Represents the total number of customer orders processed within the dataset.
- **Delayed Orders**
  Indicates the number of orders that experienced delivery delays, helping identify inefficiencies.
- **Total Stock Available**
  Shows the total inventory available across all products, supporting inventory planning.
- **Orders by Status**
  Displays the distribution of orders across different statuses, providing insight into operational flow.
- **Stock by Product**
  Highlights inventory levels for individual products, helping identify low-stock items.
- **Delay Days per Order**
  Measures the delay in delivery for each order, useful for performance evaluation.
- **Order Trend Over Time**
  Tracks how the number of orders changes over time, supporting demand analysis and forecasting.

Data Inputs

a) Source

The dataset used in this project is a **synthetic (manually created) dataset** designed to simulate real-world supply chain operations. The data was created using **Microsoft Excel** to represent typical scenarios involving order processing, inventory tracking, and shipment delivery across multiple warehouses.

The dataset was then imported into **Power BI** for data modeling and visualization.

b) Description

The dataset is divided into three main tables:

Orders Table

Contains details of customer orders.

- Order_ID
- Product_Name
- Quantity
- Order_Date
- Customer_City
- Status (Delivered, Delayed, Packed, Shipped, In Transit)
- Warehouse

Inventory Table

Contains stock information for each product.

- Product_ID
- Product_Name
- Warehouse

- Stock_Available
- Reorder_Level

Shipments Table

Contains delivery and shipment-related details.

- Shipment_ID
- Order_ID
- Dispatch_Date
- Expected_Delivery
- Actual_Delivery
- Delay_Days

These tables together provide a complete representation of supply chain operations, enabling tracking of orders, monitoring of inventory, and analysis of delivery performance.

c) Database Design

The database is structured using a **relational design approach**, where each table represents a specific component of the supply chain.

- The **Orders table** acts as the central entity
- The **Inventory table** provides stock-related information
- The **Shipments table** provides delivery-related information

This structure ensures organized data storage, reduces redundancy, and supports efficient analysis for dashboard creation.

d) Relationships – Schema

Relationships between tables are defined to integrate the data effectively:

- **Orders ↔ Shipments**
  Linked using **Order_ID**
  This allows tracking of shipment status and delay for each order
- **Orders ↔ Inventory**
  Linked using **Product_Name**
  This enables analysis of inventory levels for ordered products
- These relationships form a simple relational schema that supports data integration and analysis in Power BI.

e) DBMS, if any

- No external DBMS is used in this project.
- Data management is handled using:
- **Microsoft Excel** for dataset creation and storage
- **Power BI** for data modeling, relationship management, and visualization
- Power BI internally functions as a lightweight data management system for analysis.

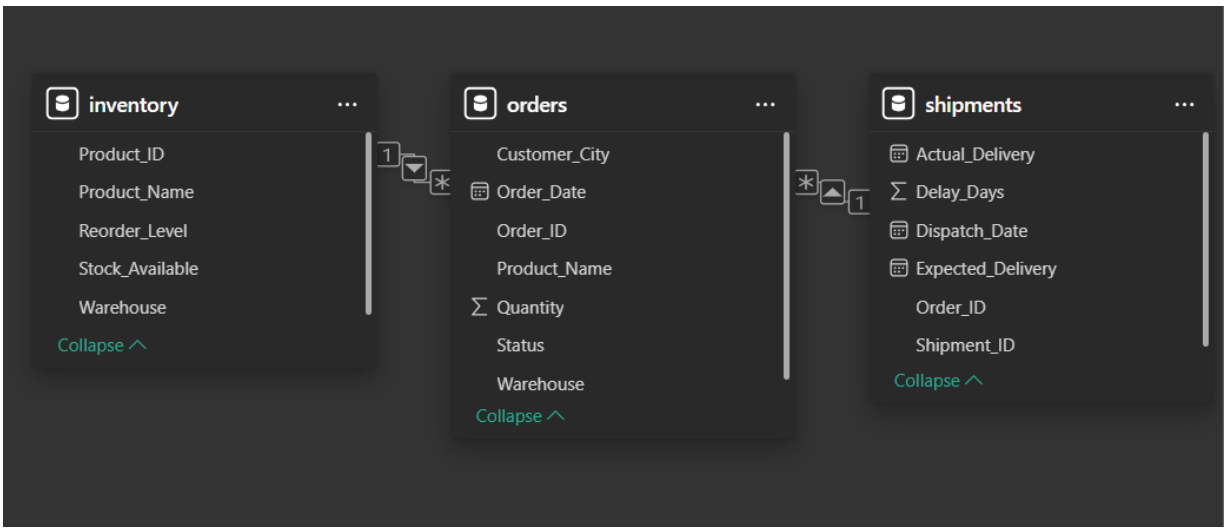

Figure 1: Data Model Showing Relationships Between Tables

Data Outputs / Visualization

Figure 2: Overall Supply Chain Dashboard

a) Dashboard Design & Graphs (Including Types of Charts)

The data outputs of this project are presented through an interactive dashboard developed using **Power BI**. The dashboard is designed to provide a clear and comprehensive view of supply chain operations, including order tracking, inventory monitoring, and shipment delay analysis.

The layout of the dashboard is structured into key sections:

1. KPI Cards

KPI cards are placed at the top of the dashboard to provide a quick summary of important metrics:

- **Total Orders** – total number of orders processed
- **Delayed Orders** – number of orders with delivery delays
- **Total Stock Available** – total inventory across all products

These indicators help users quickly understand the overall system performance.

Figure 3: Key Performance Indicators (KPIs)

2. Orders by Status (Clustered Column Chart)

This chart displays the distribution of orders across different statuses such as Delivered, Delayed, Packed, Shipped, and In Transit.
It helps identify bottlenecks and inefficiencies in the order lifecycle.

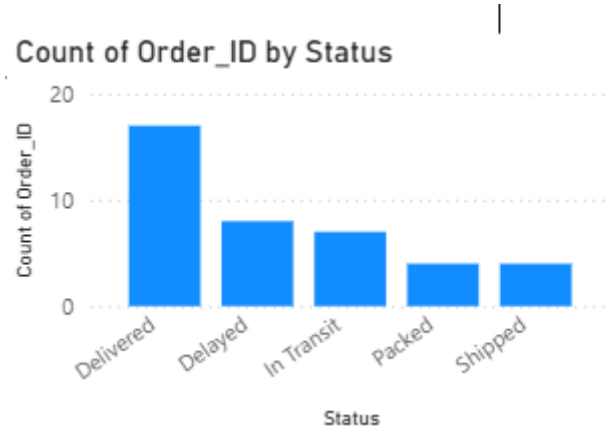

Figure 4: Order Status Distribution

3. Inventory by Product (Bar Chart)

This chart shows the stock levels of different products.
It helps identify low-stock and high-stock items, supporting better inventory management.

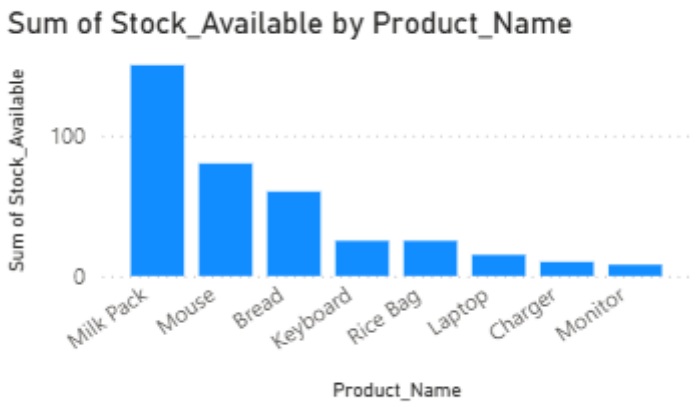

Figure 5: Inventory Stock Distribution by Product

4. Shipment Delay Analysis (Bar Chart)

This visualization displays delay days associated with each order.
It helps analyze delivery performance and identify orders that experienced delays.

5. Orders Trend by Date (Column Chart / Line Chart)

This chart shows how the number of orders changes over time.
It helps identify demand patterns and supports forecasting analysis.

b) Interactivity

The dashboard is interactive and user-friendly, allowing users to explore the data dynamically.

- A **warehouse slicer** is provided to filter data based on location (e.g., Bangalore, Chennai)
- All charts automatically update based on selected filters
- Users can perform focused analysis by selecting specific categories

This interactivity enhances usability and enables efficient decision-making.

c) Report Samples

The dashboard provides meaningful insights that can be used for reporting and analysis:

- Identification of delayed orders and delivery issues
- Monitoring of inventory levels across products
- Analysis of order distribution across different statuses
- Tracking of demand trends over time
- Warehouse-wise performance comparison

Machine Learning

a) List of ML Algorithms / Techniques Used and their Purpose

This project uses **Linear Regression**, a supervised machine learning algorithm, to perform predictive analysis on supply chain data.

1. Demand Forecasting (Order Prediction)

- **Algorithm Used:** Linear Regression
- **Purpose:** To predict the number of customer orders for the next day based on historical order data
- **Application:**
  The model is trained using past order trends (day vs number of orders) and predicts future demand.
  This helps in planning inventory levels and ensuring product availability.

Example output:

Predicted orders for next day ≈ 9.57

2. Shipment Delay Prediction

- **Algorithm Used:** Linear Regression
- **Purpose:** To estimate delay in shipment deliveries
- **Application:**
  The model uses historical delay data to predict expected delay days for future shipments.
  This helps identify potential delivery issues and improve logistics planning.

Example output:

Predicted delay for next shipment ≈ 1.14 days

These models provide predictive insights that complement the dashboard's analytical capabilities.

b) How ML Enhances the Solution

Machine learning enhances the system by adding **predictive capabilities** to the existing analytical dashboard.

- Enables forecasting of future order demand
- Helps in proactive inventory planning
- Reduces chances of stockouts or overstocking
- Identifies potential shipment delays in advance
- Supports data-driven decision-making
- Improves overall efficiency of supply chain operations

By combining visualization with prediction, the system becomes more intelligent and forward-looking.

c) ML Tools Used

The machine learning models were implemented using:

- **Platform:** Google Colab (cloud-based Python environment)
- **Programming Language:** Python

Libraries Used:

- **NumPy** – for numerical computations
- **Matplotlib** – for plotting graphs and visualization
- **Scikit-learn** – for implementing Linear Regression models

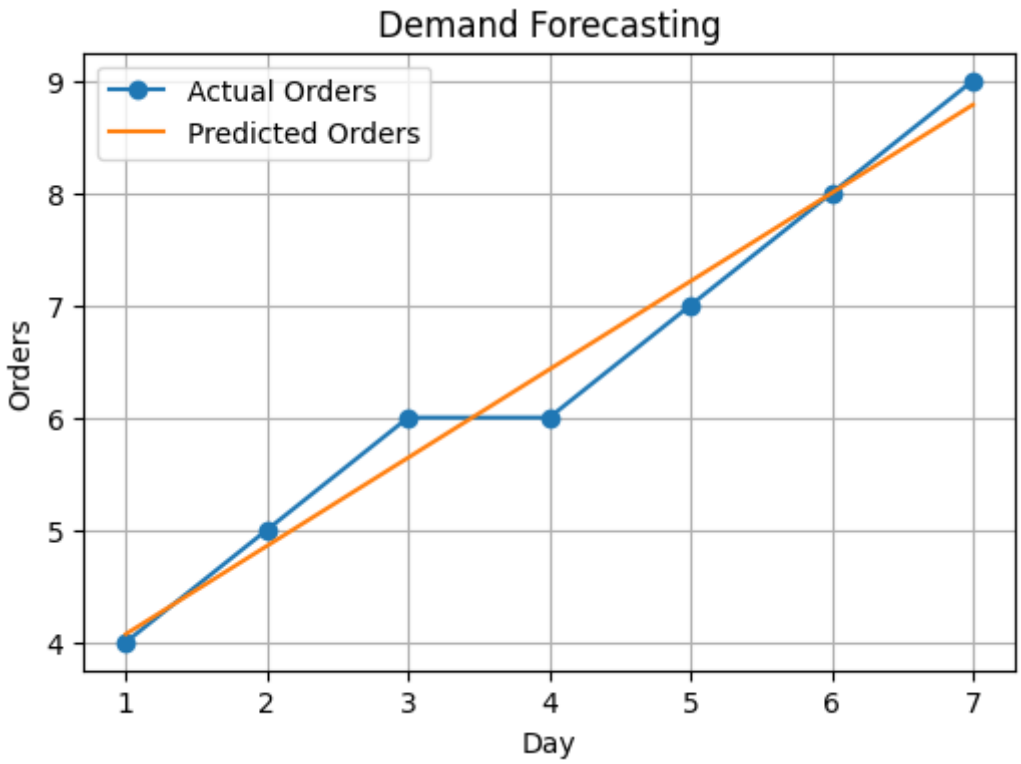

Figure 6: Demand Forecasting using Linear Regression (Actual vs Predicted Orders)

Results, Conclusion and Future Work

Summary Discussion

The project successfully demonstrates how supply chain operations can be improved using data visualization and machine learning techniques. The developed Power BI dashboard provides a clear and interactive view of key components such as order status, inventory levels, and shipment delays.

By integrating multiple datasets, the system enables efficient tracking and monitoring of supply chain activities. The dashboard allows users to quickly identify operational issues such as delayed orders and low stock conditions.

In addition, machine learning models were implemented using Linear Regression to perform demand forecasting and delay prediction. The models produced reliable predictions, showing that future order trends and potential delays can be estimated effectively using historical data.

Overall, the project combines analytical and predictive approaches to enhance supply chain visibility and decision-making.

Business Insights and Impacts

The analysis of the dataset and dashboard visualizations provides several important insights:

- A portion of orders experience delays, indicating potential inefficiencies in the delivery process
- Certain products have lower stock levels, highlighting the need for timely inventory replenishment
- Order distribution across statuses helps identify bottlenecks in processing stages
- The order trend shows relatively consistent demand, which can be used for planning and forecasting
- Warehouse-wise filtering enables comparison of performance across locations

Business Impact

- Improves visibility into supply chain operations
- Enables faster identification of delays and inefficiencies
- Supports better inventory management and stock planning
- Helps in forecasting future demand using machine learning
- Enhances decision-making through data-driven insights

Conclusion

This project demonstrates the effectiveness of combining data visualization and machine learning in solving supply chain challenges. The interactive dashboard provides real-time insights into orders, inventory, and shipment performance, while the predictive models enable forecasting of demand and delays.

The system improves operational efficiency, reduces uncertainty, and supports proactive decision-making. It highlights the importance of data-driven approaches in modern supply chain management.

Future Work

The project can be further enhanced in the following ways:

- Use real-world and larger datasets for more accurate analysis
- Implement advanced machine learning models such as Time Series or ARIMA for better forecasting
- Integrate real-time data sources for live tracking
- Develop a web-based or mobile application for wider accessibility
- Add automated alerts for low stock and delayed shipments
- Include more supply chain parameters such as supplier data and transportation costs

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